Coulomb’s law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate electric force to charge magnitude and separation.

SUCCESS CRITERIA - I can...

1. Identify charge signs before predicting direction.
2. Use inverse-square scaling.
3. Calculate force with charge in coulombs and distance in meters.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “charge” and “separation” to explain your model or result. Sentence starters are optional.

charge: a property responsible for electric interactions

separation: distance between the specified objects or centers

inverse square: proportional to one divided by distance squared

01 Warm-up: bring back an earlier idea

A wave travels at fixed speed. A second target is 3 times farther away than the first.
Compare one-way travel times.

Word help: wave = a disturbance that can carry energy and information

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F = k|q_1q_2|/r^2$; $k = 9.0\times10^9\ N\cdot m^2/C^2$.
r is separation. Signs determine attraction/repulsion; the equation gives magnitude.

Worked reasoning: Doubling the separation between two fixed charges makes Coulomb-force magnitude one quarter as large. The charge signs determine attraction or repulsion separately.

Your turn: Sketch both separations for two positive charges and label the force directions.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Label charges and separation.

Object A Object B

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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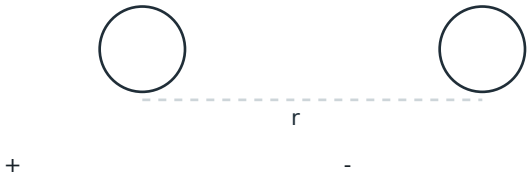
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two opposite charges are moved twice as far apart. **Draw** their force directions and compare the new force with the original.

Sentence starter: Both … . They differ in … because … .



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

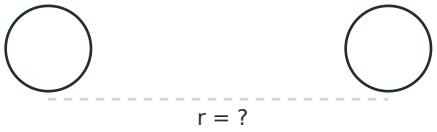
Marzano check - circle: 1 2 3

04 Apply the physics

Two fixed charges attract with a force of 12 N in a simulation. Double their center-to-center separation while keeping both charges fixed. **Label** both separations and calculate the new force magnitude using the inverse-square relationship.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Charge 1 Charge 2

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

One charge has twice the magnitude of the other. Does it exert twice the force on the smaller charge?
Explain.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 600 Hz sound travels through air. Do air particles mainly move parallel or perpendicular to propagation?
Name the wave type.
Label a wave sketch to support the calculation or comparison.

Sentence starter: The … in my model represents … . Its label shows … .



Time or position: choose and label

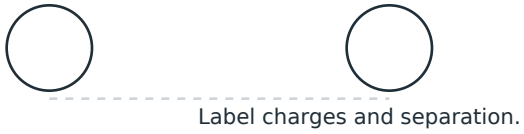
Marzano check - circle: 1 2 3

07 Exit check

With separation fixed, one charge magnitude triples. What happens to the force magnitude?

Word help: magnitude = size of a quantity without its direction

Sentence starter: I notice … . This supports … because … .



Label charges and separation.

Object A Object B

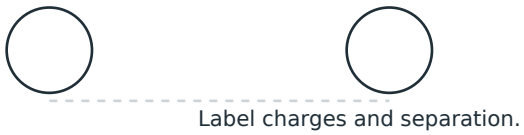
Marzano check - circle: 1 2 3

08 STRETCH: test the model

A force is 12 N. Double one charge and double separation.
Find the new magnitude and explain each factor.

Word help: magnitude = size of a quantity without its direction

Sentence starter: My claim is … . The diagram/data show …, so … .



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3